

TEAM ID	SWTID-2026-7349
PROJECT TITLE	India's Agricultural Crop Production Analysis using Tableau
MAX MARKS	5 MARKS

CHAPTER – VII

FUNCTIONAL AND PERFORMANCE TESTING

7.1 – Performance Testing & Functional Testing

Amount of Data Rendered to Tableau

The dataset used for this project contains 246,091 rows and 7 columns, with a total file size of approximately 12 MB. The dataset was successfully imported into Tableau Desktop, and all dashboards, maps, charts, and stories were rendered efficiently without noticeable performance issues.

Utilization of Filters

Multiple interactive filters were implemented across the Tableau dashboards to enable users to explore agricultural data dynamically. The following filters were used:

- State Filter: Allows users to analyse crop production data for one or more Indian states.
- District Filter: Enables district-level analysis within the selected state.
- Crop Filter: Displays information for a selected crop such as Rice, Wheat, Cotton, Maize, etc.
- Season Filter: Filters data based on agricultural seasons such as Kharif, Rabi, Summer, Winter, and Whole Year.
- Crop Year Filter: Allows users to analyse production trends for a specific year (1997–2021).

These filters are interconnected, allowing users to perform detailed drill-down analysis and compare production, area, and yield across different dimensions.

Number of Calculated Fields

In Tableau, Calculated Fields are used to create new dimensions or measures using formulas. They help derive meaningful insights from existing data without modifying the original dataset.

For this project, the following calculated fields were created:

1. Yield
 - Formula: $\text{SUM}([\text{Production}]) / \text{SUM}([\text{Area}])$
 - Purpose: Calculates crop yield per hectare.
2. Season Category
 - Categorizes crops into seasons such as Kharif, Rabi, Summer, Winter, and Whole Year.
3. Unit Type
 - Identifies the unit of production (Tonnes, Bales, or Number) based on the selected crop.

Total Number of Calculated Fields Created: 3

These calculated fields improve dashboard functionality by enabling accurate KPI calculations, seasonal analysis, and yield comparison across different crops and regions.

ALPHA
— INNOVATORS —

